

Krish Chawla

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Work Eligibility: Curricular Practical Training (CPT) - Student (F-1) Visa

EDUCATION

Master of Science, applying to begin January 2027 (program to be determined)

Bachelor of Science (GPA: 3.92), State University of New York at Fredonia (Dec 2026)

Major: Computer Science, Dual Minors: Applied Mathematics and Data Science

Alpha Lambda Delta Honor Society Member (Inducted Mar 2025)

Pi Mu Epsilon National Mathematics Honor Society (Inducted 2026)

CERTIFICATIONS

TestOut Security Pro Certification, CompTIA (Dec 2024)

How IBM Bob and AI Tools Are Changing the Way Solutions Are Built, IBM (Issued Jul 2026)

Highest Commendation Certificate, United Nations Summit on Mental Health (2020)

RESEARCH

Statistical Aging-curve Models – Independent Researcher (accepted to begin Sept 2026)

Fighter Aging Curves in the UFC | Faculty Sponsor: Dr. Megan Johnson

- Awarded the Donald Nasca Undergraduate Student Research & Creativity Award (\$2,500) for original research

TECHNICAL SKILLS

Programming Languages: Python, C, C++, R, TypeScript, SQL

Web & Development: React, Express, PostgreSQL, Prisma, Auth0

AI & Data Science: HuggingFace, YOLO, Machine Learning, Data Visualization

Tools & Platforms: Linux, GitHub, Docker, AWS, Azure, GCP, VS Code, IntelliJ, Neovim

TECHNICAL EXPERIENCE

Artificial Intelligence Intern (Jan 2026-Present)

ZubAir - Center for Innovation & Economic Development, SUNY Fredonia

- Support emerging AI initiatives involving prompt engineering and deployment of pre-trained AI models
- Collaborate with faculty and project stakeholders to explore practical AI applications and workflows

Traffic Analytics Data Programming Intern (June-Aug 2024)

ZubAir - Center for Innovation & Economic Development, SUNY Fredonia

- Analyzed large-scale traffic datasets to identify vehicle usage patterns and congestion trends
- Utilized Linux, Python, YOLO, and Supervision libraries for data collection and analysis
- Created data visualizations and optimized database reporting processes
- Recorded and evaluated vehicle speed metrics to support transportation analyses

Robotics Project Team Member – Capstone (Sept-Dec 2024)

Computer Science Department, SUNY Fredonia

- Configured robotic systems and assisted with testing in the Robotics Laboratory under faculty supervision

TECHNICAL PROJECTS

Jirassic Park - Agile Task Management Platform

React / TypeScript / Express / PostgreSQL / Prisma / Auth0 (github.com/burk3405/Jirassic-Park)

- Built backend REST API for task CRUD, sharing (VIEW/EDIT permissions), and advanced search/filtering
- Implemented Auth0 JWT authentication middleware and CORS configuration; debugged a Prisma ORM WHERE-clause precedence bug affecting task ownership filtering
- Wrote unit test suite achieving 27/27 passing tests
- Applied the Repository design pattern via Prisma ORM for type-safe, SQL-free data access

TECHNICAL PROJECTS, continued...

Rubik's Cube Solver (Mar-May 2026)

C/C++

- Implemented Korf's Algorithm (IDA* search with Pattern Database heuristics) to solve the Rubik's Cube optimally across its $\sim 4.3 \times 10^{19}$ -state space
- Built pattern database heuristic tables and a benchmarking harness to empirically validate solve time and node expansion against theoretical complexity bounds
- Confirmed pattern-database-guided search drastically reduced nodes expanded relative to naive breadth-first search, consistent with theoretical expectations

Predicting UEFA Champions League Success (Apr 2026)

Python / Machine Learning / Scikit-Learn

- Engineered per-90 performance metrics from 6 merged Kaggle/UEFA 2021–22 datasets covering goals, attacking, defending, and goalkeeping
- Applied K-Means clustering to identify 4 player archetypes and Random Forest (5-fold CV) to predict match outcomes from team-level stats
- Found goal-scoring output as the strongest predictor of match success, with pass accuracy as a multiplying factor
- Presented findings at the SURC West Research Conference (2026)

Mathematical Contest in Modeling (MCM) Team Member (Mar 2026)

Faculty Advisor: Lan Cheng, SUNY Fredonia

- Solved MCM Problem C ("Data with The Stars") with a 3-person team, modeling unknown fan vote estimates from Dancing with the Stars judges' scores and contestant data across 34 seasons
- Compared rank-based and percent-based vote-combination methods to evaluate fairness of the show's elimination system
- Designated a Successful Participant

Non-Preemptive CPU Scheduling Simulation (Jan-May 2026)

Python

- Implemented FCFS, SJF, and Priority Scheduling algorithms in an object-oriented, tick-based simulation
- Ran controlled experiments (10 trials $\times$ 15 processes) comparing algorithm performance on waiting time and turnaround time
- Found Priority Scheduling outperformed FCFS by $\sim 37\%$ in average waiting time (136.7 vs. 217.7 ticks)

LEADERSHIP & TEACHING

Peer Tutor - Computer Science & Mathematics (100+ students weekly) | SUNY Fredonia (Aug 2025-Present)

- Completed Tutor Theory & Practice certification training for documentation

Teaching Assistant - Programming Languages | SUNY Fredonia (Feb 2024-Present)

- Conduct tutoring sessions, academic support activities, and assist with coursework and assignment review

Computer Lab Proctor | SUNY Fredonia | February 2024-Present

- Monitor laboratory operations and provide technical assistance to students and faculty

GENERAL EMPLOYMENT

Fitness Center Desk Attendant, Blue Devils Fitness Center, SUNY Fredonia (Aug 2024–Present)

Desk Attendant (Substitute), E-Sports Lounge, SUNY Fredonia (Oct 2023–Present)

Food Service Attendant, Cranston Marche Dining Hall, SUNY Fredonia (Jun-Aug 2024)

STUDENT ENGAGEMENT & CULTURAL PARTICIPATION

Chess Club, SUNY Fredonia (Aug 2023–Present)

Commencement Event Volunteer, SUNY Fredonia (May 2024)

Student Presenter Outreach & Admissions, International Student Services, SUNY Fredonia (Apr 2024)

Multicultural Weekend Visitor Host – Hosted 4 High School students, SUNY Fredonia (Mar 2024)